1-2 Evaluation of Uncertainty

Source / 來源

- 選修物理(一)課堂講義（4623R52）教用版，第1章 不確定度，§1-2 不確定度的評估，pp. 7–9

- Halliday, Resnick & Walker, *Fundamentals of Physics*, Ch 1: Measurement

- Young & Freedman, *University Physics with Modern Physics*, Ch 1: Units, Physical Quantities, and Vectors

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Examples

Core Ideas — Type-A and Type-B Evaluation

For n repeated measurements of X,

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad SD = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}}$$

The Type-A standard uncertainty is obtained statistically:

$$u_A(X) = \frac{SD}{\sqrt{n}}$$

Type-B evaluation uses instrument resolution, published uncertainty, or other non-statistical information. With only a smallest scale division d and an assumed uniform distribution,

$$u_B(X) = \frac{d}{2\sqrt{3}}$$

Independent contributions combine in quadrature:

$$u(X) = \sqrt{u_A(X)^2 + u_B(X)^2}$$

Report a result as $X \pm u(X)$. Keep at most two significant figures in $u(X)$, rounding upward when the next digit is nonzero, and round the best estimate to the same decimal place.

Scientific notation makes significant figures explicit: $302.06 = 3.0206 \times 10^2$ has five significant figures, while $0.00205 = 2.05 \times 10^{-3}$ has three.

Worked Example — Standard Uncertainty *(p. 8; 範例說明)*

A stopwatch has a smallest scale division of $0.01\ \mathrm{s}$. A steel ball is dropped from $1.50\ \mathrm{m}$ and the fall time is measured five times: 0.58, 0.57, 0.61, 0.63, and $0.58\ \mathrm{s}$. How should the result be reported?

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Example 1 — Uncertainty *(p. 9; 範例 1)*

Atai uses a balance with a $0.1\ \mathrm{kg}$ smallest division; Xiaoyu uses one with a $1\ \mathrm{kg}$ smallest division. They measure the same fruit basket four times.

<table>
<tr><th>Measurement</th><th>Atai (kg)</th><th>Xiaoyu (kg)</th></tr>
<tr><td>1</td><td>16.77</td><td rowspan="2">16.8</td></tr>
<tr><td>2</td><td>16.75</td></tr>
<tr><td>3</td><td>16.77</td><td>16.7</td></tr>
<tr><td>4</td><td>16.78</td><td>16.8</td></tr>
<tr><td>Mean</td><td>16.7675</td><td>16.775</td></tr>
<tr><td>SD</td><td>0.01258</td><td>0.05</td></tr>
</table>

Find each reported result and decide whose result is more trustworthy.

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Related Problem 1 — Pendulum Period *(p. 9; 類題 1)*

Ming measures a pendulum period four times: 1.3, 1.2, 1.3, and $1.4\ \mathrm{s}$. The calculator gives $\bar{x} = 1.3000\ \mathrm{s}$ and $SD = 0.08165\ \mathrm{s}$. The stopwatch resolution is $0.1\ \mathrm{s}$. Which report is correct?

(A) $1.300 \pm 0.029\ \mathrm{s}$

(B) $1.300 \pm 0.041\ \mathrm{s}$

(C) $1.300 \pm 0.050\ \mathrm{s}$

(D) $1.300 \pm 0.081\ \mathrm{s}$

(E) $1.300 \pm 0.10\ \mathrm{s}$

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Quick Check

Q1 — Interpreting a Package Label

A package is labelled “Weight: $195 \pm 15\ \mathrm{gw}$.” If the label follows the ISO uncertainty convention, which statement is correct?

(A) 195 is one sampled measurement.

(B) ± 15 is the standard uncertainty.

(C) ± 15 is the best estimate.

(D) The weight must vary between 180 and $210\ \mathrm{gw}$.

(E) With Type-B uncertainty only, the smallest scale division may be $15\ \mathrm{gw}$.

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Q2 — Two Stopwatches *(select two)*

Student	Five readings (s)	Mean (s)	SD (s)
Ming	0.5893, 0.5812, 0.5868, 0.5836, 0.5874	0.5857	0.0032
Hua	0.58, 0.58, 0.58, 0.58, 0.58	0.58	0.00

Ming's stopwatch resolves $0.0001\ \mathrm{s}$; Hua's resolves $0.01\ \mathrm{s}$. Which statements are correct? *(Select two.)*

(A) Ming: $0.5857 \pm 0.0029\ \mathrm{s}$

(B) Ming: $0.5857 \pm 0.0015\ \mathrm{s}$

(C) Hua: $0.58 \pm 0.00\ \mathrm{s}$

(D) Hua: $0.5800 \pm 0.0029\ \mathrm{s}$

(E) Hua's uncertainty is smaller.

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Exam Archive

Q1 — Jewelry Scale *(2023, 112 分科)*

A jeweler measures a gold ornament five times with a balance whose smallest division is 1 mg. The mean is $m_{AV} = 95.367823\ \mathrm{g}$ and $u_C = 0.35686524\ \mathrm{mg}$. Which statement is correct?

(A) $u_A = SD$

(B) $u_B = 1\ \mathrm{mg}$

(C) $u_C = (u_A + u_B)/2$

(D) Report $m_{AV} = 95.4\ \mathrm{g}$ and $u_C = 0.4\ \mathrm{mg}$.

(E) Report $m_{AV} = 95.36782\ \mathrm{g}$ and $u_C = 0.36\ \mathrm{mg}$.

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Glossary

English term	中文	Definition / Note
Type-A evaluation	A 類評估	Statistical evaluation from repeated measurements
Type-B evaluation	B 類評估	Evaluation using resolution, references, or other information
sample standard deviation	樣本標準差	Spread of a finite data sample
standard uncertainty	標準不確定度	Combined uncertainty $u(X)$
instrument resolution	儀器精密度／最小刻度	Smallest distinguishable increment
uniform distribution	均勻分布	Equal probability over a stated interval
quadrature	平方和原則	$\sqrt{u_1^2 + u_2^2 + \cdots}$
best estimate	最佳估計值	Rounded central value in the reported result
significant figures	有效數字	Digits that meaningfully represent a value
measurement result	測量結果	Best estimate written with its uncertainty